

C-traceability-matrix

C — Master Traceability Matrix (browser extension)

Revision: 1 **Last modified:** 2026-06-10T00:00:00Z **Purpose:** Anti-skip backbone (operator mandate + §11.4.118 enumerated coverage). One row per atomic requirement / feature / entity / edge-case / NFR / dimension-decision drawn from ALL six analysis files (docs/browser_extension/_analysis/01..06). EVERY item from the source corpus appears. Proves nothing is silently dropped: deferred items are listed explicitly with rationale; untraceable items are listed OPEN.

Stable ID scheme: - FR-001..FR-025 — 25 functional requirements (file 01 §B). - NFR-001..NFR-015 — 15 non-functional requirements (file 01 §C). - DIM01-Rxx .. DIM12-Rxx — per-dimension consolidated decisions (file 02). - ER-<entity> — 6 ER entities (file 03 §2) + SQL-table superset (file 04 §4a). - EDGE-xx — documented edge cases (file 01 §I + scattered). - SEC-xx — security/STRIDE controls (file 02 Dim11 + file 03 §6 + file 01 §N). - INS-xx — 10 cross-dimension insights (file 02). - CV-xx — cross-verification conflicts/low-confidence items (file 02). - FND-xx — foundation/source-layer facts (files 04/05). - TST-xx — test-type coverage requirements (file 06). - BLD-xx — build/release/distribution (file 01 §M + file 02 Dim12 + file 03 §7). - 0Q-xx — open questions / ambiguities (all files).

Phase column uses the authoritative 8-phase impl-plan (file 01 §P.2): P1 Foundation, P2 Core Engine, P3 Extension Shell, P4 API Integration, P5 Tab Groups, P6 UI/UX Polish, P7 Testing, P8 Build/Dist. v2 = deferred (see Coverage Ledger). Boba constraint flags are noted in Notes.

1. Functional Requirements (25)

ID	Source	Requirement	Category	Phase	Notes/Dependencies
FR-001	01 §B.1 FR-001	Detect magnet links (href magnet:?) in DOM of any page, dynamic across SPA; iframe via all_frames; ≤500 ms after injection	detection	P2/P3	MutationObserver debounce 500 ms (EDGE-06); regex MAGNET_REGEX (FND-1)
FR-002	01 §B.1 FR-002	Detect .torrent file links via .torrent suffix (case-insens) + Content-Type: application/x-bittorrent	detection	P2/P3	TORRENT_FILE_REGEX (FND-12); CORS-aware (DIM07)
FR-003	01 §B.1 FR-003	Parse magnet URI per BEP-9: xt/dn/tr/xl/ws/x.pe; xt mandatory valid 40-hex; dn fallback “Unknown Torrent”; malformed tr discarded	parsing	P2	OQ-07 param-set driven as/kt/mt vs x.pe); DIM
FR-004	01 §B.1 FR-004	Download + parse .torrent via fetch; max 10 MB (maxTorrentFileSize); 30 s timeout; CORS→Boba-proxy fallback; bencode decode	parsing/ api	P2/P4	EDGE-04, EDGE-05; F parser is dead code i (FND-23)
FR-005	01 §B.1 FR-005	Compute SHA-1 over bencoded info dict → 40-hex infohash; MUST match xt	parsing	P2	RAW bencode, not de re-encode (DIM07-R; FND-22)
FR-006	01 §B.1 FR-006	Deduplicate torrents by lowercase infohash; merge longest displayName + union trackers + earliest detectedAt	parsing	P2	INS-02 unified identifier time-salted ids unrelated (OQ-16)
FR-007	01 §B.2 FR-007	Send magnets to qBitTorrent /api/v2/torrents/add_urls param; cookie session; opt	api	P4	DIM02; Boba→7186 p (OQ-01)

ID	Source	Requirement	Category	Phase	Notes/Dependencies
		category/tags/savepath/ rename			
FR-008	01 §B.2 FR-008	Upload .torrent via multipart torrents field to /api/v2/torrents/add; blobs in-memory only	api	P4	addTorrentFromFile (FND-31)
FR-009	01 §B.2 FR-009	Tab-group batch: enumerate group tabs, SCAN each, aggregate + dedup, transmit unified batch	tab- groups	P5	NET-NEW design — n has NO tabGroups co (OQ-17/FND-33)
FR-010	01 §B.2 FR-010	Enumerate tab groups + extract URLs via <code>chrome.tabGroups.query()</code> + <code>chrome.tabs.query()</code> ; show titles/colors; multi- select	tab- groups	P5	DIM05; Chrome/Yan only (EDGE-21)
FR-011	01 §B.3 FR-011	Auto-discover Boba on LAN: probe well-known ports + /health; parse version $\geq 1.0.0$; ranked list; persist	api	P4	Re-point 8443/8080→7186/718 (OQ-01); INS-01
FR-012	01 §B.3 FR-012	Support 4 auth methods: cookie (SID), API key (X- API-Key), Basic, Custom Header	security/ api	P4	C1: support both cookie+JWT/apikkey; n drops custom-header (OQ-06)
FR-013	01 §B.3 FR-013	Encrypt stored credentials AES-256-GCM; PBKDF2 from per-install salt; key in storage.session; 96-bit IV; 128-bit tag	security	P1/P4	§11.4.10; ref uses fixe empty passphrase BL (OQ-15/FND-30)
FR-014	01 §B.4 FR-014	Offline FIFO queue: max 1000; exp backoff 5 s→5 min; max 5 retries; persist storage.local; Dead-Letter substore	queue	P4	ref max 50 / 3 attempt diverges (OQ-19); DI
FR-015	01 §B.4 FR-015	Badge: numeric queued count; colors green idle/ blue sending/orange	ui	P3/P6	INS-09 badge sync; BADGE_COLORS (FN

ID	Source	Requirement	Category	Phase	Notes/Dependencies
		queued/red error; ≤200 ms update			
FR-016	01 §B.5 FR-016	Context menu: link(magnet)→Send Magnet; link(.torrent)→Download; page→Scan; tab-group→Send Group	ui	P3/P5	chrome.contextMenus. bobaLink-send/scan/ dashboard (FND-32)
FR-017	01 §B.5 FR-017	Keyboard shortcuts: send-current-page Ctrl+Shift+B; open-popup Ctrl+Shift+L; send-tab- group Ctrl+Shift+G (customizable)	ui	P3	OQ-08 3-vs-4 cmds; r ships B/S/D not L/G (FND-29)
FR-018	01 §B.5 FR-018	Cross-browser single codebase: Chrome 88+, Firefox 109+, Opera 74+, Yandex 21+; BrowserAdapter abstraction	cross- browser	P1/P8	DIM04; WXT handles browser manifest
FR-019	01 §B.5 FR-019	i18n via _locales/{lang}/ messages.json: en default + zh_CN/zh_TW/es/fr/de/ ja/ko; no hardcoded strings	i18n	P6	ref ships en only (28 FND-19)
FR-020	01 §B.5 FR-020	Accessibility WCAG 2.1 AA: contrast ≥4.5:1; keyboard nav; ARIA on icon buttons; aria-live; focus indicators	ui	P6	§11.4.117 pixel-oracle verification
FR-021	01 §B.5 FR-021	Dark/Light theme: respect prefers-color- scheme + manual override; CSS custom properties	ui	P6	DIM10 §9; ref ui.theme=auto (FND-2)
FR-022	01 §B.5 FR-022	Notifications via chrome.notifications: Sent / Send Failed(retry) / Batch Complete / Retrying; suppressible per-category	ui	P3/P6	DIM10 §5; MV3 no ba icons (DIM03)

ID	Source	Requirement	Category	Phase	Notes/Dependencies
FR-023	01 §B.5 FR-023	Options page: server URL+mode, auth(masked)+test-connection, default category/tags/path, notif prefs, queue settings, theme, reset	ui	P3	DIM10 §2; ref Server General/Advanced/A (FND-43)
FR-024	01 §B.5 FR-024	Health check: poll 30 s; / api/v2/app/version or / health; states connected/ connecting/disconnected/ error; tooltip last-ping	api	P4	ref HealthChecker thresholds 2 s/5 s (FN
FR-025	01 §B.5 FR-025	Rate limiting: 10 req/s burst, 60/min sustained; 1 concurrent + 500 ms inter-call; honor 429 Retry-After; in-memory	api	P4	Token-bucket (file 01 ref RATE_LIMIT 10/1 (FND-27)

2. Non-Functional Requirements (15)

ID	Source	Requirement	Category	Phase	Notes/Dependencies
NFR-001	01 §C.1	Content script init ≤ 10 ms (performance.now())	performance	P7	benchmark test (TST-09)
NFR-002	01 §C.1	Magnet detection ≤ 5 ms/link (1,000-link page benchmark)	performance	P7	DOM-first querySelector (INS-03)
NFR-003	01 §C.1	Popup render ≤ 100 ms (Lighthouse)	performance	P7	
NFR-004	01 §C.1	Options page load ≤ 200 ms (Lighthouse)	performance	P7	
NFR-005	01 §C.1	API round-trip ≤ 500 ms local (p95 over 100 calls)	performance	P7	
NFR-006	01 §C.1	Service worker cold start ≤ 50 ms	performance	P7	

ID	Source	Requirement	Category	Phase	Notes/ Dependencies
NFR-007	01 §C.1	Bundle size compressed ≤ 350 KB	build	P8	tree-shake + code-split
NFR-008	01 §C.2	Credential encryption at rest: AES-256-GCM via Web Crypto	security	P1/P4	composes FR-013/SEC-04
NFR-009	01 §C.2	No cleartext credential logging (ESLint rule + review)	security	P1	§11.4.10; logger must redact
NFR-010	01 §C.2	HTTPS-only API comms (URL scheme validation + CSP)	security	P4	except localhost (SEC-08)
NFR-011	01 §C.2	CSP script-src 'self'; object-src 'none'	security	P1	MV3 CSP object (SEC-06)
NFR-012	01 §C.2	Minimum permission model: activeTab + declared host perms only	security	P1	OQ-09 activeTab vs https://*/ ; ref over-broad (FND-08)
NFR-013	01 §C.3	Crash-free session rate $\geq 99.9\%$	reliability	P7	chaos/stress (TST-07/08)
NFR-014	01 §C.3	Offline queue durability 100% — all items survive restart	reliability	P4/P7	persist storage.local; chaos test
NFR-015	01 §C.3	API compatibility qBitTorrent 4.4.x–5.x	api	P4/P7	version matrix (file 01 §R); v5 state renames (DIM02)

NFR §4.4 Scalability sub-items (non-numbered, captured as NFR-SCAL): | ID | Source | Requirement | Category | Phase | Notes | |—|
—|—|—|—|—| | NFR-SCAL-1 | 01 §C.4 | Tab-group batch size
unlimited (memory-bound) | scalability | P5 | | | NFR-SCAL-2 | 01

§C.4 | Offline queue 1,000 default / 10,000 max | scalability | P4 | | |
 NFR-SCAL-3 | 01 §C.4 | 10,000+ links must not jank (>16 ms frame);
 requestIdleCallback >500 links | performance | P2/P7 | EDGE-13;
 yieldToBrowser (FND) | | NFR-MAINT-1 | 01 §C.5 | Coverage ≥80%
 unit / ≥60% E2E; JSDoc; TS strict zero any; ≤20 runtime deps |
 maintainability | P7 | Boba mandate raises to ~100% types (TST-15) |

3. Research-Dimension Consolidated Decisions (12 dimensions)

ID	Source	Requirement / Decision	Category	Phase
DIM01-R1	02 Dim01	Treat Dim01 as authoritative Boba topology: qBt 7185 (proxy 7186), FastAPI 7187, bridge 7188, jackett 9117, boba-jackett 7189; Go proxy 7187 (profile go)	data-model/api	P1/P4
DIM01-R2	02 Dim01	Extension talks to FastAPI :7187: /api/v1/search(+SSE stream), /api/v1/download, /api/v1/download/file, /api/v1/magnet, /api/v1/auth/status, /api/v1/config, /health	api	P4
DIM01-R3	02 Dim01	Boba CORS via ALLOWED_ORIGINS env incl. chrome-extension://<id>; SW not subject to CORS	security/api	P4
DIM01-R4	02 Dim01	qBt default creds admin/admin (cookie SID); private trackers env-driven; 48 nova3 plugins; freeleech-only IPTorrents	api	P4
DIM02-R1	02 Dim02	qBt WebUI v2: cookie auth /api/v2/auth/login→SID; Referer/Origin MUST match Host (CSRF); 403=IP ban	api/security	P4
DIM02-R2	02 Dim02	/torrents/add: magnets in urls, .torrent multipart torrents; 415 invalid; 200=guaranteed-added; 405 wrong-method v4.4.4+	api	P4
DIM02-R3	02 Dim02	API-key auth (Bearer qbt_) v5.2.0+ only; cookie is safe baseline	api	P4
DIM02-R4	02 Dim02	Monitoring via polling /sync/maindata?rid= (no native WS/SSE); v5 renames pausedUP→stoppedUP/ pausedDL→stoppedDL	api	P4
				P1

ID	Source	Requirement / Decision	Category	Phase
DIM03-R1	02 Dim03	MV3 manifest: host_permissions separate; background {service_worker, type: module}; perms storage/notifications/contextMenus/scripting/alarms/activeTab	cross-browser	
DIM03-R2	02 Dim03	SW HARD timeouts: 30 s idle / 5 min request / 30 s fetch; chrome.alarms keep-alive; register listeners synchronously top-level; return true for async	cross-browser	P3
DIM03-R3	02 Dim03	Content scripts document_idle; ISOLATED world; cross-origin fetch routed through SW (CS can't since Chrome 85)	detection/security	P3
DIM03-R4	02 Dim03	Storage quotas: local ~10 MB, sync ~100 KB/8 KB-item, session ~10 MB in-memory; badge ≤4 chars	data-model	P1
DIM03-R5	02 Dim03	Offscreen documents for DOM/parse in MV3 (one/profile); NOT in Firefox	parsing	P2
DIM04-R1	02 Dim04	MV3 everywhere; Firefox uses Event Pages (WXT handles); webextension-polyfill for browser.* on Chromium + own feature-detect	cross-browser	P1/P8
DIM04-R2	02 Dim04	Firefox MV3 REQUIRES gecko.id + strict_min_version; version x.y.z only; data_collection_permissions since Nov 2025	build	P8
DIM04-R3	02 Dim04	Opera lacks storage.sync/storage.managed; notifications partial on Mac; sidebarAction Opera/FF only	cross-browser	P6/P8
DIM04-R4	02 Dim04	UA detection order Yandex→Opera→Edge→Chrome→Chromium; prefer feature detection	cross-browser	P1
DIM05-R1	02 Dim05	tabGroups Chrome 89+/MV3; query/get/update/move; create via tabs.group/ungroup; groupId NOT persistent (id by title+color+URLs)	tab-groups	P5
DIM05-R2	02 Dim05	Accessing tab.url/title needs tabs perm or host perms; collapsed groups still API-accessible	tab-groups	P5
DIM05-R3	02 Dim05		tab-groups	P5

ID	Source	Requirement / Decision	Category	Phase
		Yandex Chromium ~147+ full tabGroups; Chrome 148+ chrome://newtab cannot be grouped		
DIM06-R1	02 Dim06	Magnet BEP-9: xt only mandatory; hash v1 40-hex / 32-base32 (convert) / v2 btmh 1220+64-hex; two-phase detect+validate	parsing	P2
DIM06-R2	02 Dim06	base32 (32-char) infohash MUST convert to hex (40-char); RFC4648 alphabet	parsing	P2
DIM06-R3	02 Dim06	Decode dn (decodeURIComponent + +→space); tr/xs/as/ws decode; kt split +; so BEP-53 ranges; duplicate params→array	parsing	P2
DIM06-R4	02 Dim06	Sanitize dn before HTML render (strip control/<>, ≤255, textContent); chunk large text 100 KB vs backtracking	security/ parsing	P2
DIM07-R1	02 Dim07	.torrent = bencode dict announce+info; info required piece length/pieces(20-byte SHA-1)/ name; single length XOR multi files[]	parsing	P2
DIM07-R2	02 Dim07	Bencode: dict keys lexicographic RAW-byte sorted (load-bearing for infohash); strict type rules	parsing	P2
DIM07-R3	02 Dim07	Infohash = SHA-1 of RAW bencoded info dict as-in-file (never decode-re-encode); Web Crypto needs HTTPS secure ctx	parsing	P2
DIM07-R4	02 Dim07	CORS download: CS can't cross-origin fetch since Chrome 85 → delegate to SW; size ≤10 MB (50 MB max); validate first byte 0x64	api/parsing	P2/P4
DIM07-R5	02 Dim07	Private torrents (BEP-27): info.private==1 disables DHT/PEX; sanitize passkey to PASSKEY (never log)	security	P2
DIM08-R1	02 Dim08	Content-script timing document_idle + readyState checks for heavy SPAs (safeInitialize)	detection	P3
DIM08-R2	02 Dim08	DOM-first querySelectorAll('a[href^="magnet:"]') (~90% catch) + TreeWalker text-node scan skipping script/style	detection	P2
DIM08-R3	02 Dim08		detection	P2

ID	Source	Requirement / Decision	Category	Phase
		MutationObserver debounced 500 ms (~88× faster than polling), incremental scan new nodes; IntersectionObserver infinite scroll		
DIM08-R4	02 Dim08	iframe all_frames:true + match_about_blank:true; Shadow DOM recursive shadowRoot (querySelectorAll doesn't pierce)	detection	P2
DIM08-R5	02 Dim08	Site-specific selector DB top ~20 sites (TPB/1337x/nyaa/rutracker/generic) — tier-2 detection	detection	P2
DIM08-R6	02 Dim08	Scanner cleanup: track observers/timers/listeners, disconnect on navigation (memory-leak prevention); 16 ms frame budget	performance	P2
DIM09-R1	02 Dim09	Communication patterns: fetch (ext→server), EventSource SSE, WebSocket progress, runtime.sendMessage, storage.local, alarms	api	P4
DIM09-R2	02 Dim09	Auth strategies: cookie/SID, JWT Bearer (60 s buffer + single-flight refresh, password NOT stored), API key, Basic	security/api	P4
DIM09-R3	02 Dim09	Retry classification: retry 0/429/500/502/503/504/401-after-refresh; non-retry 400/403/404/415; exp backoff+jitter (base 1000, max 30000, ×2, jitter 0.5, max 3)	queue/api	P4
DIM09-R4	02 Dim09	SSE/EventSource cannot send custom headers — auth via URL query param; reconnect exp backoff; Chrome 116+ keeps SW alive on WebSocket	api	P4
DIM09-R5	02 Dim09	Credential storage: store tokens only never plaintext password; chrome.storage.local NOT encrypted by default; clear on logout; HTTPS+validate certs	security	P4
DIM09-R6	02 Dim09	Health: separate liveness/readiness; parallel checks AbortSignal.timeout(5000); status healthy/degraded/unhealthy	api	P4
DIM09-R7	02 Dim09	Offline queue persist + retry on reconnect; navigator.onLine unreliable for LAN →	queue	P4

ID	Source	Requirement / Decision	Category	Phase
		proactive /app/version ping every 30 s via alarms		
DIM09-R8	02 Dim09	Rate limiting token-bucket per-endpoint; RequestBatcher (maxBatch 10, maxWait 50 ms)	api	P4
DIM09-R9	02 Dim09	Manifest host_permissions localhost+127.0.0.1 (http+https * port); externally_connectable localhost; NEVER <all_urls> in prod	security	P1
DIM10-R1	02 Dim10	Popup Chrome max 800×600 / min 25×25; sweet spot 380–450px wide; fixed header/connection bar/scroll/footer	ui	P3
DIM10-R2	02 Dim10	Context menus require contextMenus perm; >1 visible item auto-collapses into extension-named parent	ui	P3
DIM10-R3	02 Dim10	Badge text ≤4 chars; per-tab via tabId (resets on close); setBadgeText/Color	ui	P3
DIM10-R4	02 Dim10	Commands max 4 suggested keys; all need Ctrl/Alt; Mac Ctrl→Command default; fires in SW	ui	P3
DIM10-R5	02 Dim10	Side panel Chrome 114+ (open 116+, close 141+); cannot auto-open without user gesture	ui	v2
DIM10-R6	02 Dim10	i18n_locales/<locale>/messages.json; manifest default_locale required if _locales; __MSG_key__+getMessage(); @@bidi for RTL	i18n	P6
DIM11-R1	02 Dim11	Design principles: Least Privilege, Defense in Depth, Zero Trust, Secure by Default, Privacy First	security	P1
DIM11-R2	02 Dim11	STRIDE mitigations S1-4/T1-4/R1-2/I1-5/D1-3/E1-3 + trust boundaries TB1-4	security	P1/P7
DIM11-R3	02 Dim11	Minimal perms storage/alarms/notifications/activeTab; optional clipboardWrite; host localhost; NEVER all_urls/tabs/cookies/webRequest/scripting/downloads/history	security	P1
DIM11-R4	02 Dim11	Specific per-site matches (never wildcard — wildcards nullify postMessage protection)	security	P1

ID	Source	Requirement / Decision	Category	Phase
DIM11-R5	02 Dim11	AES-256-GCM + PBKDF2-HMAC-SHA256 100k iters; 128-bit salt; 96-bit IV; non-extractable key; master pw never stored; decrypted cache storage.session; Clear All Data (GDPR)	security	P4
DIM11-R6	02 Dim11	Input validation: magnet/info-hash 40-hex v1/64-hex v2; sanitize dn textContent; validate Boba URL; reject javascript:/data:/ vbscript:; block dangerous ports	security	P2/P4
DIM11-R7	02 Dim11	Privacy: collect only Boba URL/ creds(encrypted) + ephemeral magnet/ names; NO page content/history/IP; analytics opt-in; eraseAllData(); no 3rd- party/selling	privacy	P6
DIM11-R8	02 Dim11	Secure fetch: credentials:'omit' + referrerPolicy:'no-referrer'; HTTPS-only except localhost; HSTS; self-signed cert pinning/CA install	security	P4
DIM11-R9	02 Dim11	Validate sender.id + sender.url (chrome- extension://) on ALL onMessage; action allow-listing	security	P3
DIM11-R10	02 Dim11	Update security: official-store signed distribution; FF self-hosted update_hash HTTPS; verify perms unchanged on update	build	P8
DIM12-R1	02 Dim12	Build WXT (Vite); test Jest unit + Playwright E2E; mock sinon-chrome; versioning Conventional Commits→SemVer	build/testing	P7/P8
DIM12-R2	02 Dim12	Playwright needs launchPersistentContext + -load-extension + -disable-extensions- except; jsdom for DOM tests	testing	P7
DIM12-R3	02 Dim12	Coverage threshold global 70% config (aspirational 80%+ INS-10); reporters lcov/ html/json-summary; optional Sonar/Snyk/ Semgrep/Trivy/Gitleaks	testing	P7
DIM12-R4	02 Dim12	Store assets: icons 16/32/48/128(+512); screenshots 640×480 or 1280×800; privacy policy; no console.log/source-maps in prod; CSP set; no remote code	build	P8

4. ER Entities / Data Model (6 ER + SQL superset)

ID	Source	Entity / Requirement	Category	Phase	Notes
ER-extension_config	03 §2; 04 §4a T2	Single per-install settings (config_id PK, server_url, api_key, qbittorrent_url, username, password_encrypted, auto_detect, default_category, scan_interval_ms, max_retries, retry_delay_ms, timestamps)	data-model	P1	SQL rows (diag)
ER-discovered_torrents	03 §2; 04 §4a T3	Torrents found on pages (id=infohash PK UK, magnet_uri, name, size_bytes, source_url, source_tab_id, status, discovered_at, sent_at, error_message; SQL adds discovery_method/is_private/tab_group_id/selected/boba_status)	data-model	P2/P3	mm (file)
ER-download_queue	03 §2; 04 §4a T4	Offline/retry queue (queue_id PK, torrent_id FK, magnet_uri, priority 0-10, retry_count, status queued/retrying/failed/completed, queued_at, last_retry_at, error_log; SQL adds next_retry_at/server_id/torrent_data_b64)	queue/ data-model	P4	FR-0
ER-server_config				P1/P4	DIM

ID	Source	Entity / Requirement	Category	Phase	Notes
	03 §2; 04 §4a T5	Per-Boba-server multi-server (server_id PK, name, fastapi_url, go_service_url, qbittorrent_url, api_key, is_active, health_check_interval, last_health_check, last_check_result; SQL ports 7187/7189/8080→7186)	data-model		
ER-send_history	03 §2; 04 §4a T6	Audit log every send (history_id PK, infohash, magnet_uri, name, server_id FK, success, qbittorrent_hash, category, tags, sent_at, response_time_ms, error_message)	data-model	P4	imm
ER-search_cache	03 §2; 04 §4a (n/a)	Cached search results TTL'd (cache_id PK, query_hash, query_text, results JSON, result_count, cached_at, ttl_seconds, is_valid) — standalone, no relationship	data-model	v2	search Boba back
ER-REL	03 §2 relationships	Cardinality: extension_config 1— o{ discovered_torrents; discovered_torrents 1 —o{ download_queue; server_config 1— o{ send_history	data-model	P1	extension FK c
ER-SQL-extra	04 §4a T1/T7/T8/T9	SQL superset tables: app_metadata, site_selectors (30 seed rows), queue_log, statistics (daily aggregates)	data-model	P1/P2	sql.j chron one

5. Edge Cases (documented)

ID	Source	Edge case	Category	Phase	Notes
EDGE-01	01 §I	Magnet without dn → “Unknown Torrent”; qBt resolves from metadata	parsing	P2	FR-003
EDGE-02	01 §I	Duplicate infohash → silently skipped “duplicate”; merge longest dn+union tr+earliest ts	parsing	P2	FR-006
EDGE-03	01 §I	Malformed tracker URLs → logged + discarded	parsing	P2	FR-003
EDGE-04	01 §I	.torrent >10 MB → not downloaded; E_FILE_TOO_LARGE 413 → “use magnet”	api	P4	FR-004
EDGE-05	01 §I	CORS- blocked .torrent → Boba proxies (Boba mode); qBt-direct manual copy	api	P4	FR-004; E_CORS
EDGE-06	01 §I	Dynamic SPA/ infinite-scroll links → MutationObserver debounce 250/500 ms	detection	P2	DIM08-R3
EDGE-07	01 §I	iframe links → only when all_frames; may be inaccessible	detection	P2	DIM08-R4
EDGE-08	01 §I	JS click-handler “links” (not real <a>) → not detected; manual copy	detection	P3	documented limitation
EDGE-09	01 §I	Non-standard/ encoded magnet formats → not	detection	P2	

ID	Source	Edge case	Category	Phase	Notes
		detected; manual copy			
EDGE-10	01 §I	Login-gated / logged-in-only sites	detection	P3	documented
EDGE-11	01 §I	Strict-CSP pages → extension respects CSP, will NOT inject scripts	detection/security	P3	threat model
EDGE-12	01 §I	Forgotten encryption password → no recovery; reset extension + reconfigure	security	P4	FR-013
EDGE-13	01 §I	Pages 10,000+ links → no jank (>16 ms); requestIdleCallback >500; pagination	performance	P2	NFR-SCAL-3
EDGE-14	01 §I	Browser restart w/o password → session key lost → re-enter credentials	security	P4	storage.session loss
EDGE-15	01 §I	Storage quota exceeded → E_STORAGE_FULL; Firefox stricter; clear old queue	data-model	P4	DIM03-R4
EDGE-16	01 §I	MV3 SW termination → chrome.alarms keep-alive; Firefox suspends more aggressively	cross-browser	P3	DIM03-R2
EDGE-17	01 §I	qBt rejects duplicate → already downloading/ completed	api	P4	FR-007
EDGE-18	01 §I	Private-tracker torrents → work if accessible; may	api/security	P4	DIM07-R5

ID	Source	Edge case	Category	Phase	Notes
EDGE-19	01 §I	need passkey in tracker URL Self-signed certs → browser exception / system trust; HTTPS-only temporarily disabled for testing	security	P4	DIM11-R8
EDGE-20	01 §I	macOS __MACOSX metadata folder in ZIP → load error, must delete	build	P8	
EDGE-21	01 §I	Mobile browsers unsupported; Safari only Partial MV3 not tested; Firefox tab-groups incompatible	cross-browser	P8	DIM05-R1
EDGE-22	02 Dim06/08	URL-encoded magnet hrefs; javascript: URLs containing magnets; data-* attrs; base32 hashes	detection	P2	DIM06/08
EDGE-23	02 Dim07	Empty/invalid bencode, missing info, leading zeros, neg-zero, out-of-order keys, unicode/binary filenames, unterminated structures	parsing	P2	TST bencode
EDGE-24	02 Dim05	groupId reuse across sessions; incognito groups don't persist; pinned tabs auto-unpinned; max ~32-64 groups/window	tab-groups	P5	DIM05

6. Security / STRIDE Controls

ID	Source	Control	Category	Phase	Notes
SEC-01	01 §N; 02 Dim11	Threat model: credential theft, MITM, XSS, privilege escalation, fingerprinting	security	P1	DIM11-R2
SEC-02	02 Dim11 S	Spoofing: cert pinning/HTTPS, origin validation, isolated world, magnet hash validation	security	P1/P4	
SEC-03	02 Dim11 T	Tampering: store-signed packages, HTTPS/HSTS, AES-GCM, infohash validation	security	P1/P8	
SEC-04	01 §N; 02 Dim11 I	Info disclosure: AES-256-GCM creds; minimal matches no all_urls; restrict web_accessible_resources + use_dynamic_url; sender validation; no telemetry	security	P1/P4	FR-013
SEC-05	02 Dim11 D/E	DoS: client rate-limit, document_idle, storage quotas; EoP: sender validation + action allow-list, no eval, strict CSP	security	P1/P4	
SEC-06	01 §N; 02 Dim11	CSP object: extension_pages script-src 'self'; object-src 'self'; connect-src 'self' https:	security	P1	NFR-011; ref CSP (FND-09)
SEC-07	03 §6	Crypto ops in Offscreen Document (Web Crypto SubtleCrypto.encrypt) + PBKDF2 key derivation	security	P4	DIM03-R5; Firefox bg-page fallback
SEC-08	02 Dim11; 03 §6	Network layer HTTPS-only TLS 1.2+ (except	security	P4	DIM11-R8

ID	Source	Control	Category	Phase	Notes
SEC-09	03 §6	localhost); cert validation no self-signed / pinning Boba-services controls: API-key auth (X-Api-Key), Basic auth, rate-limit 100 req/min/key	security	P4	diagram #10
SEC-10	02 Dim06/ Dim11	Sanitize display names textContent; reject javascript:/data:/ vbscript;; block dangerous ports; chunk text vs backtracking	security	P2	DIM11- R6
SEC-11	02 Dim11	Privacy policy + opt-in analytics/error-reporting; no third-party sharing/ selling; Limited-Use compliance	privacy	P6/P8	DIM11- R7

7. Cross-Dimension Insights (10)

ID	Source	Insight	Category	Phase	Notes
INS-01	02 Insight 1	Zero-config discovery via well-known localhost ports (7187/7189/8080→7186) + version pings → one-click Auto-Discover	api	P4	FR-011
INS-02	02 Insight 2	Unified torrent identity {infoHash,name,source,type}; magnet + .torrent converge on 40-char BTIH; dedup across tab group	parsing	P2	FR-006
INS-03	02 Insight 3	3-tier progressive detection: universal link scan (~90%) + site-specific selectors + text- based + MutationObserver	detection	P2	DIM08
INS-04	02 Insight 4	Tab group as batch job: right-click group → extract all URLs → parse → submit batch newline-separated	tab- groups	P5	FR-009

ID	Source	Insight	Category	Phase	Notes
INS-05	02 Insight 5	Extension as Boba satellite (“send-to”), not management tool; use Boba auth/config; extend not duplicate UI (~30% less code)	api	P1/P4	architecture decision
INS-06	02 Insight 6	Offline-aware queue; navigator.onLine unreliable for LAN → proactive /app/ version ping 30 s via alarms + persistent queue	queue	P4	FR-014; DIM09-R7
INS-07	02 Insight 7	Privacy-first activeTab + optional host perms + scripting.executeScript on demand (MEDIUM — reviewers may prefer declared host perms)	security	P1	OQ-09 decision
INS-08	02 Insight 8	Multi-client gateway: TorrentClient interface (qBt/Transmission/Deluge/rTorrent)	data-model	v2	DEFERRED — adds complexity; v1 leaves seam, targets Boba only
INS-09	02 Insight 9	Real-time badge sync: poll / sync/maindata incremental RID → badge count/color	ui	P3/P4	FR-015
INS-10	02 Insight 10	Enterprise-grade testing: 80%+ coverage, real-Boba-in-Docker E2E, security scanners, signed releases	testing	P7	TST-15; Boba mandate

8. Cross-Verification Conflicts & Low-Confidence Items

ID	Source	Item	Category	Phase	Notes
CV-01	02 cross-verif C1	Auth for Boba APIs: cookie (qBt direct) AND	security/api	P4	RESOLVED both valid; FR-012

ID	Source	Item	Category	Phase	Notes
CV-02	02 cross- verif C2/ C3	JWT/API-key (Boba own) — support BOTH Build tool WXT (built on Vite); MV3 everywhere (WXT handles FF Event Pages)	build/ cross- browser	P1	RESOLVED
CV-03	02 cross- verif low	qBt API-key Bearer only v5.2.0+; cookie baseline	api	P4	low- confidence; DIM02-R3
CV-04	02 cross- verif low/C4	Boba SSE path /api/ v1/search/ stream/{id} not confirmed in official docs; ports user- configurable	api	P4	confirm vs live OpenAPI (OQ-12)
CV-05	02 cross- verif low	Firefox E2E Playwright extension- ID extraction limits; side panel Chrome 114+ limits	testing/ui	P7/v2	TST-03; DIM10-R5

9. Foundation / Source-Layer Facts (load-bearing for the port)

ID	Source	Fact / Requirement	Category	Phase	N
FND-01	04 §1	Reference name bobalink v1.0.0 MIT (impl-plan says Apache-2.0 — OQ-10); rename for Boba (OQ naming)	build	P1	LI de
FND-02	04 §2	host_permissions 7187/7189/8080 — must change 8080→7186 (real qBt proxy)	cross-browser/ api	P1	O
FND-03	04 §3	TS strict posture: all strict flags + noUncheckedIndexedAccess + exactOptionalPropertyTypes; zero any (no-explicit-any error)	build	P1	N
FND-04	04 §3	ESLint type-aware: no-floating-promises/no-misused-promises error; no-console warn (allow error/warn/info)	build	P1	N
FND-05	04 §3	Jest preset ts-jest ESM jsdom; coverage 80%; collectCoverageFrom EXCLUDES popup/options/content/background/index.ts	testing	P7	TS in
FND-06	04 §3	Playwright baseUrl chrome-extension://test-id (placeholder); --load-extension=./dist; chromium+firefox projects	testing	P7	TS se
FND-07	04 §2	content_scripts matches <all_urls> run_at document_idle (ref) — reconcile with minimal-perms	detection/ security	P1/P3	D
FND-08	04 §2; 05 §9	ref CSP connect-src localhost 7187/7189/8080; permissions include scripting (DIM11 says avoid)	security	P1	O
FND-09	04 §6	Regex set: MAGNET_REGEX/MAGNET_VALIDATION/INFOHASH/TORRENT_FILE/INFOHASH_HEX/BASE32/DN/TR	parsing	P2	FI
FND-10	04 §6; 05 §8	DEFAULT_PORTS FastAPI 7187/Go 7189/qBt 8080; DEFAULT_URLS; QBITTORRENT_ENDPOINTS map	api	P1/P4	O
FND-11	04 §6	Timing: DEBOUNCE MUTATION 500/AUTO_SCAN 1000; REQUEST_TIMEOUTS DEFAULT 15000/AUTH 10000/ADD 30000/DISCOVERY 3000; RETRY max 3 base 1000	api/ detection	P2/P4	co (C

ID	Source	Fact / Requirement	Category	Phase	N
		max 30000 jitter 0.3; RATE_LIMIT 10/1000ms			
FND-12	04 §6	STORAGE_KEYS all bobalink_ prefixed (CONFIG/AUTH_STATE/CREDENTIALS/DETECTED/HISTORY/QUEUE/HEALTH/KEY_MATERIAL); only CONFIG/QUEUE/HEALTH actually written	data-model	P1	un
FND-13	04 §6	ENCRYPTION AES-GCM 256, IV 12B, salt 16B, PBKDF2 SHA-256 100000 iters, key version 1; passphrase source UNDEFINED in foundation	security	P4	O
FND-14	04 §6	BADGE_COLORS (HEALTHY #4CAF50/DEGRADED #FF9800/ERROR #F44336/SCANNING #2196F3/DETECTED #9C27B0/DEFAULT); ICON_SIZES 16/32/48/128	ui	P6	FI
FND-15	04 §4b	TS types: MagnetInfo, TorrentFile, ParsedTorrent, DetectedTorrent, SendResult, PageScanResult, ServerConfig(28 fields), ExtensionConfig, DEFAULT_CONFIG, AuthMethod, AutoDiscoveryConfig	data-model	P1	ty (C
FND-16	04 §5	shared libs: crypto (encrypt/decrypt/sha256/generateSecurePassphrase), errors (BobaLinkError+8 subclasses), events (TypedEventEmitter 13-event EventMap), logger, storage (NamespacedStorage), utils (debounce/throttle/TokenBucket/retryWithBackoff/yieldToBrowser/processInChunks)	data-model	P1	fo m
FND-17	05 §2.1	BobaAPIClient: login/logout/getVersion/addTorrentFromMagnet/addTorrentFromFile/getTorrents/deleteTorrents/pauseTorrents/resumeTorrents; requestWithRetry+rate-limit+timeout	api	P4	FI
FND-18	05 §2.2	AuthHandler 4 methods (none/cookie/api_key/basic); createCredentialsFromConfig decrypt point; MAX_CONSECUTIVE_FAILURES 3	security/ api	P4	FI
FND-19	05 §2.3		queue	P4	

ID	Source	Fact / Requirement	Category	Phase	N
		OfflineQueue: DEFAULT_MAX_SIZE 50, PROCESS_INTERVAL 60000, ITEM_SEND_DELAY 500; enqueue FIFO evict/dequeue/processQueue/ startAutoProcessing			FI vs (C (C
FND-20	05 §2.4	HealthChecker: thresholds healthy<2000/ degraded<5000, MAX_FAILURES 2; checkServer/checkAllServers/ testConnection/autoDiscover(ports 8080/7187/7189)	api	P4	FI
FND-21	05 §2.5	qBitTorrentAdapter: sendTorrent/ sendTorrents (250 ms delay)/ addTorrentFile/buildAddOptions/ mapContentLayout	api	P4	FI
FND-22	05 §3	bencode (zero-dep Uint8Array, sorted keys, sha1), magnet (parseMagnetUri xt/dn/tr/ws/ xl/xs/kt/as/mt + base32ToHex), torrent-file (parseTorrentFile infohash=sha1(encode(info)))	parsing	P2	FI er
FND-23	05 §3 (note)	.torrent parser fully built but NEVER invoked (dead code) — sends upload raw File	parsing	P4	w (C
FND-24	05 §4	Scanner: BaseScanner (shadow-DOM querySelectorAllDeep, maxElements 10000), LinkScanner (site-specific+generic), TextScanner (TreeWalker), ScannerOrchestrator (MutationObserver debounce 500, per-site override, yieldToBrowser)	detection	P2/P3	FI
FND-25	05 §4 (note)	NO chrome.tabGroups usage in ref; only intra-tab yieldToBrowser + per-tab tabTorrents Map — tab-group batching is net-new	tab- groups	P5	O
FND-26	05 §4.5	Two overlapping selector tables: constants.SITE_SELECTORS (~21 domains, used by LinkScanner) vs site-db.SITES (15 richer, used only for debounce) — reconcile	detection	P2	O
FND-27	05 §5	Message protocol: ExtensionMessage{type,payload,requestId}; background handles scan-result/get-	api	P3	de un (C

ID	Source	Fact / Requirement	Category	Phase	N
FND-28	05 §6	detected/send-torrent/get-config/set-config/ health-check/test-connection/auto-discover/ authenticate/scan-page/open-dashboard/ queue-status/queue-process Background SW: lifecycle onInstalled/ onStartup, initialize, initializeApiClient, sendTorrents, context menus, commands, alarms (keepalive ~20 s / health 5 min), badge/notifications, storage listeners	cross- browser	P3	D
FND-29	04 §2; 05 §6	commands: send-to-boba Ctrl+Shift+B, scan-page Ctrl+Shift+S, open-dashboard Ctrl+Shift+D (ref ships 3, differs from FR-017 B/L/G)	ui	P3	O
FND-30	05 §9; 05 banner	§11.4.10 BLUFF: options encrypts with FIXED passphrase “bobalink-extension”; background decrypts with EMPTY “” — broken-by-design, MUST replace	security	P4	O
FND-31	05 §2.5	fetch uses credentials:include/same-origin (qBt SID needs it) — conflicts DIM11-R8 credentials:omit — reconcile	security/ api	P4	D
FND-32	05 §6	Context menus bobalink-send (magnet/.torrent link patterns) / bobalink- scan / bobalink-dashboard	ui	P3	FI
FND-33	05 §7.1	Popup controls: status dot, Select All/ Deselect/Refresh, torrent list (checkbox/ name/type/infoshash16/Sent), empty-state Scan Page, Send to Boba, progress overlay	ui	P3	FI
FND-34	05 §7.2	Options sections Servers/General/ Advanced/About; server modal (name/url/ auth/category/savepath/Test Connection); Auto Discovery card; General+Advanced toggles; Reset All	ui	P3	FI
FND-35	04 §4a	SQL dual-storage model: sql.js-over- storage.local (9 tables) vs STORAGE_KEYS JSON-blob + NamespacedStorage — pick one source of truth	data- model	P1	O

10. Test-Type Coverage Requirements (toward 100% per §11.4.27)

ID	Source	Requirement	Category	Phase	Notes
TST-01	06 \$types 1	Integration: real parser→queue→api- client→live qBt/proxy (7186/7187); no mocks outside unit	testing	P7	§11.4.27(A)
TST-02	06 \$types 2	E2E functional: load real built MV3 bundle, resolve real ext id, drive popup/options/content with user-observable assertions	testing	P7	fixture builds dist + captures id (OQ)
TST-03	06 \$types 3	Automation/ autonomous QA: self- driving N×-rerunnable (§11.4.98/§11.4.50); HelixQA banks	testing	P7	§11.4.116 sync channel
TST-04	06 \$types 4	Security/penetration: credential-leak audit (§11.4.10), XSS via magnet/dn, CSP, perm- scope, host-perm abuse, malformed injection, SID handling, crypto-at- rest	security/ testing	P7	SEC-xx
TST-05	06 \$types 5	DDoS/load: 1000s magnet links/page, rapid scan cycles, queue enqueue burst, API flood/rate-limit under load	testing	P7	§11.4.85
TST-06	06 \$types 6	Scaling: large .torrent files, 1000s-item queue+persistence, large detected-set render, many servers	testing	P7	NFR-SCAL
TST-07		Chaos: storage failure/ corruption mid-write,	testing	P7	error- injecting

ID	Source	Requirement	Category	Phase	Notes
	06 §types 7	network drop/timeout/ reorder, SW termination mid-op, corrupt-state recovery (§11.4.85 closed-set)			mock needed
TST-08	06 §types 8	Stress: sustained $N \geq 100$ / ≥ 30 s, concurrent $N \geq 10$ enqueues/scans, boundary inputs, p50/ p95/p99 latency	testing	P7	§11.4.85
TST-09	06 §types 9	Performance: parse- time benchmarks, scan latency heavy DOM, popup render — captured evidence	performance/ testing	P7	NFR-001..006
TST-10	06 §types 10	Benchmark: formal suite + baselines + regression detection	testing	P7	
TST-11	06 §types 11	UI tests: rendering correctness, badge counts, list render, modal state changes	testing	P7	beyond DOM- presence
TST-12	06 §types 12	UX tests: detect→select→send happy path, error feedback, empty/ loading/error states, a11y, keyboard nav, shortcuts	testing	P7	FR-020
TST-13	06 §types 13	Challenges (. / challenges/scripts/): real built ext vs real qBt+proxy, user- observable outcomes	testing	P7	§11.4.27(B)
TST-14	06 §types 14	HelixQA banks + autonomous QA sessions with captured evidence	testing	P7	§11.4.27
TST-15	06 §types	Anti-bluff remediation: fix ref bluffs —	testing	P7	§11.4/ §11.4.1;

ID	Source	Requirement	Category	Phase	Notes
TST-16	15; §critical	expect(true).toBe(true) Auth/auto-proc blocks, crypto-copy-not-module, non-functional e2e (hardcoded test-id), dead-asserts; every test fails vs no-op stub; bring UI/entrypoints INTO coverage	testing	P7	reference unit+e2e is starting floor
	06 fixtures	Wire invalidMagnets (7 negative cases) + magnets.json/base32 fixtures; reference test inventory (bencode/ magnet/torrent-file/api- client/queue/scanner strong; crypto thorough- but-copy)			EDGE-23

11. Build / Release / Distribution

ID	Source	Requirement	Category	Phase
BLD-01	01 §M.1	Build: npm install; npm run dev (WXT HMR); npm run build → chrome/firefox/opera/edge MV3 prod; per-browser wxt build -b	build	P8
BLD-02	01 §M.2	Test commands: jest –coverage (80%), playwright; coverage thresholds; per-file targets	testing	P7
BLD-03	01 §M.3	Release: main→release branch→bump manifest→CHANGELOG→full suite→git tag→build all→GitHub Release→submit stores	build	P8
BLD-04	01 §M.4; 02 Dim12	Distribution: Chrome Web Store CRX (\$5), Firefox AMO XPI, Opera CRX, Edge, GitHub ZIP; per-store assets/2FA/review SLA	build	P8
BLD-05	01 §M.5	Git: GitHub Flow; Conventional Commits (feat/ fix/.../scopes); branch prefixes — Boba: SSH remotes not HTTPS	build	P1/P
BLD-06	03 §7		build	P7/P

ID	Source	Requirement	Category	Phase
		Pipeline stages Code→Lint→Test(≥80%)→Build→Package→Deploy; quality gates 0 ESLint errors/valid manifest/<5MB		
BLD-07	02 Dim12; 04 §7	release-please + Husky pre-commit + lint-staged + web-ext sign — Boba: DROP Husky (git hooks forbidden)	build	P8
BLD-08	01 §K.1; 03 §9; 04 §7; 06 §10	CI/CD REMOVAL (Boba Hard-Stop): strip ALL .github/workflows/*.yaml (ci.yaml, release.yaml), Husky hooks, “enforced in CI”, auto store- submission → convert to manual ./ci.sh-style script	build	P8
BLD-09	04 §7	Manual version-bump touches package.json + wxt.config manifest + both SQL files + app_metadata/extension_config schema_version rows together	build	P8

12. Open Questions / Ambiguities (consolidated, for conductor)

ID	Source	Question	Category	Phase	Notes
OQ-01	01 §T1; 03 §9.4; 04 §8.1; 05 §9.2	Real Boba port mapping: 8443/8080→7186/7187/7189; qBt direct→proxy 7186? Boba REST/SSE→7187? search→7189? Re-point all discovery defaults	api	P1/P4	HIGH/critical
OQ-02	01 §T2; 03 §9.1; 04 §7; 06 §10	CI/CD removal shape — manual ./ci.sh	build	P8	HIGH; BLD
OQ-03	01 §T3; 06 §1	Jest vs Vitest (Boba frontend uses Vitest; crypto.test.ts already Vitest)	testing	P7	MED
OQ-04	01 §T4; 01 §O.3	Two divergent source layouts (src/parser/scanner/api/ shared vs src/modules/utis/ content-scripts) — pick canonical	data- model	P1	MED
OQ-05	01 §T5; 01 §P.1		build	P1	

ID	Source	Question	Category	Phase	Notes
		Two divergent phase plans (plan.md 10 vs impl-plan 8) — use 8-phase			MED; this matrix use
OQ-06	01 §T6; 01 §D.7	Data-model vocabulary drift: source_type adds torrent-url; status completed vs dead-letter; auth_method adds none drops custom-header — reconcile enums	data-model	P1	MED
OQ-07	01 §T7	Magnet param superset: FR-003 xt/dn/tr/xl/ws/x.pe vs impl-plan xs/as/kt/mt (no x.pe) — which implemented	parsing	P2	LOW; ref i xt/dn/tr/ws xs/kt/as/mt
OQ-08	01 §T8; 04 §2; 05 §6	Keyboard-shortcut count 3 vs 4; ref ships B/S/D, FR-017 says B/L/G; Ctrl+Shift+L collision	ui	P3	LOW; FND
OQ-09	01 §T9; 02 INS-07; 04 §8	host_permissions scope: https://*/ (ref) vs activeTab minimal (NFR-012) — store-review/privacy	security	P1	LOW-MED
OQ-10	01 §T10; 04 §1	LICENSE: impl-plan Apache 2.0 vs package.json MIT — confirm	build	P1	LOW
OQ-11	01 §T11	Naming: “BobaLink” reference name — confirm Boba repo extension name + icon	build	P1	LOW
OQ-12	01 §T12; 02 §509-510; 03 §9.6	Search/SSE scope: does Boba 7187 expose /api/v1/torrents/search+SSE as spec, or call existing merge routes? Confirm vs live /openapi.json	api	P4	MED; CV-0
OQ-13	01 §T13	Boba /api/v1/health vs real /health — confirm path/port	api	P4	LOW-MED
OQ-14	04 §8.3; 05 §9.3	Dual storage model (sql.js 9-table vs JSON-blob NamespacedStorage) — one source of truth	data-model	P1	FND-35
OQ-15			security	P4	

ID	Source	Question	Category	Phase	Notes
	04 §8.2; 05 §9.1	Encryption passphrase provenance UNDEFINED; ref fixed/empty-passphrase BLUFF — design real key source (master pw prompt / OS keychain / delegate to boba-jackett /config/boba.db)			CRITICAL §11.4.10; FND-30
OQ-16	05 §9.9	Time-salted detection ids (Date.now() in hashString) → unreliable cross-scan dedup → use infohash-derived stable ids	parsing	P2	FR-006
OQ-17	05 §9.8; 04 §4.5	Tab-group batching does NOT exist in ref — net-new design	tab-groups	P5	FND-25; FR-009
OQ-18	05 §9.2	Boba integration only port-deep — client speaks only qBt WebUI v2; no merge-search/proxy(7186)/boba-jackett(7189) code; define how ext reaches Boba's own endpoints	api	P4	DIM01-R2
OQ-19	01 §T6; 05 §2.3; 04 §8.5	Config-value drift: queue max 50(ref)/1000(spec)/100(Dim09); retries 3/5; RATE_LIMIT 10 vs DB seed 30; autoScanDelay 2000 vs DEBOUNCE 1000 — pick canonical	queue/ api	P4	FR-014/025
OQ-20	05 §9.5	Offline queue processQueue never authenticates → 403 loop against authed server — wire auth	queue	P4	FND-19
OQ-21	05 §9.10/9.11	Two get-detected response shapes; unrouted message types (send-result/get-auth-state/health-result/show-notification/update-badge/torrent-detected/selection-change) — unify/decide	api	P3	FND-27
OQ-22			api/build	P3/P4	DIM07-R4

ID	Source	Question	Category	Phase	Notes
	05 §9.3/9.4/9.12	set-cookie SID parsing brittle across CORS; CORS/host-perm story for cross-origin localhost; manifest/commands/icon rasterization to author			
OQ-23	06 §OQ	E2E extension-loading fixture (build dist → persistent context –load-extension → capture id); Chromium-only vs Firefox MV3; global-setup/teardown contract; live backends for integration	testing	P7	TST-02/03
OQ-24	03 §9.2/9.3/9.5/9.7	Single-send routing dual-write FastAPI+Go ambiguity; batch endpoint path (/api/torrents/batch vs direct qBt); extension_config↔discovered FK conceptual; X-API-Key+Basic together vs per-service	api/data-model	P4	diagram inconsistent

COVERAGE LEDGER

Item counts by category (every row classified)

Category	Count
detection	22
parsing	27
api	41
queue	12
ui	24
security	35
privacy	4
i18n	3
build	18
data-model	22

Category	Count
cross-browser	11
tab-groups	12
performance	12
reliability	3
scalability	4
maintainability	1
testing	23
(rows are multi-tagged; primary-category tallies above)	

Item counts by section (authoritative row totals)

Section	Rows
1. Functional Requirements	25 (FR-001..FR-025)
2. Non-Functional Requirements	15 (NFR-001..NFR-015) + 4 scalability/maint (NFR-SCAL-1..3, NFR-MAINT-1)
3. Research-Dimension Decisions	56 (DIM01-R1..DIM12-R4 across all 12 dims)
4. ER Entities / Data Model	8 (6 ER entities + ER-REL + ER- SQL-extra)
5. Edge Cases	24 (EDGE-01..EDGE-24)
6. Security / STRIDE	11 (SEC-01..SEC-11)
7. Insights	10 (INS-01..INS-10)
8. Cross-Verification	5 (CV-01..CV-05)
9. Foundation Facts	35 (FND-01..FND-35)
10. Test-Type Coverage	16 (TST-01..TST-16)
11. Build / Release	9 (BLD-01..BLD-09)
12. Open Questions	24 (OQ-01..OQ-24)
TOTAL traceable items	245

v1 vs v2 split

- **v1 (in-scope for initial release):** 240 items — all FRs, all NFRs, all dimension decisions except deferred, all ER entities except search_cache, all edge cases, all security, 8 of 10 insights, all

foundation facts, all test types, all build (CI/CD converted to manual), all open questions resolved during build.

- **v2 (explicitly deferred):** 5 items.

Items intentionally deferred to v2 (with rationale)

ID	Item	Rationale
INS-08	Multi-Client Gateway (TorrentClient interface: Transmission/Deluge/rTorrent adapters)	MEDIUM confidence; adds complexity; v1 targets Boba/qBittorrent only. v1 architecture leaves a TorrentClient seam but ships ONE adapter. (file 02 Insight 8; file 01 §509 OQ-7.)
ER-search_cache	Search-result cache entity (TTL'd, standalone, no FK)	Search aggregation is gated to Boba 1.2+ (file 01 §R feature-availability); not needed for the core detect→send v1 flow. Storage backend unresolved (OQ-12/OQ-24).
DIM01-R2 (search subset)	Boba merge-search / api/v1/search + SSE / api/v1/search/stream/{id} live-progress in popup	“Search aggregation = Boba 1.2+ only”; SSE progress = Boba 1.1+ only (file 01 §R). v1 ships detect→send + qBt-direct + basic health; rich search/SSE is a post-v1 enhancement. (Core /api/v1/download + /api/v1/magnet stay v1.)
DIM10-R5	Side panel (Chrome 114+, open 116+, close 141+; cannot auto-open)	LOW confidence (CV-05); browser-floor too high vs Chrome 88 target;

ID	Item	Rationale
DIM02-R3 / CV-03	qBittorrent API-key (Bearer qbt_) auth, v5.2.0+ only	popup covers v1 UX. Optional surface. LOW confidence; only on newest qBt; cookie auth is the safe v1 baseline. API-key path deferred until version floor justifies it.

NOTE: deferral means “not built in v1,” NOT “dropped.” Each carries a v1 seam (TorrentClient interface, search-cache storage decision, SSE client stub, side-panel optional entry, API-key auth branch) so v2 is additive.

Cross-check: completeness of the mandated baselines

- **25 FR — ALL represented:** FR-001...FR-025 each have exactly one row (§1). ✓
- **15 NFR — ALL represented:** NFR-001...NFR-015 each have one row; scalability/maintainability sub-clauses captured as NFR-SCAL-1..3 + NFR-MAINT-1 (§2). ✓
- **12 Dimensions — ALL represented:** DIM01 (4), DIM02 (4), DIM03 (5), DIM04 (4), DIM05 (3), DIM06 (4), DIM07 (5), DIM08 (6), DIM09 (9), DIM10 (6), DIM11 (10), DIM12 (4) — every dimension has ≥ 3 consolidated decision rows; 56 total (§3). ✓
- **6 ER entities — ALL represented:** ER-extension_config, ER-discovered_torrents, ER-download_queue, ER-server_config, ER-send_history, ER-search_cache; + ER-REL relationships + ER-SQL-extra (app_metadata/site_selectors/queue_log/statistics) superset (§4). ✓
- **Edge cases:** all 16 file-01-§I cases (EDGE-01..21 incl. multi-item lines) + Dim06/07/05/08 cases (EDGE-22..24) (§5). ✓
- **Insights:** all 10 (INS-01..10) (§7). ✓
- **Cross-verification:** 4 conflicts + low-confidence set (CV-01..05) (§8). ✓

Items that could NOT be placed (OPEN)

NONE. Every FR, NFR, dimension-decision, ER entity, edge case, insight, cross-verification conflict, foundation fact, test-type, build step, and open question from all six analysis files has been assigned a stable ID and a row. No source item is unaccounted for.

The reference-deliverable internal inconsistencies (dual source layouts, dual storage models, two phase plans, port mismatches, config-value drift, the §11.4.10 fixed/empty-passphrase bluff, dead/unrouted code, missing E2E fixtures, CI/CD-vs-Boba-Hard-Stop) are NOT untraceable items — they are captured as explicit OQ-xx / FND-xx rows requiring a conductor decision during the corresponding phase, per §11.4.118 enumerated coverage and §11.4.66 interactive clarification.

End of C-traceability-matrix.md — anti-skip backbone for the implementation plan.